Fault-tolerant Control Allocation for Multirotor Helicopters using Parametric Programming

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***Abstract*—This paper addresses the issue of flight safety and reliability of unmanned multirotor helicopters. New applications involve the use of such flying platforms to perform inspection of industrial facilities, bridges, dam walls, etc. In this context, the multirotor helicopter is teleoperated by a distant human user. [1] The capability of fault-tolerance against rotor or motor is a major asset to guarantee the success of the mission and the safety of the surrounding environment. This paper contributes to fault-tolerant flight of multirotor helicopter with the following four points. First, given the configuration of the multirotor vehicle, this paper shows how to calculate the maximum torques that can be produced on each vehicle axis. Second, a method to investigate the static controllability of the vehicle in any arbitrary fault configuration is described. Third, a control allocator for multirotor systems is developed with performance going beyond performance obtained with classical methods, such as the pseudo-inverse method. The proposed control allocator is indeed capable of exploiting the whole attainable control set. Fourth, this control allocator is fault-tolerant, as it can efficiently cope with the loss of one or more actuators.**

The control allocation problem is formulated as a parametric program and solved for an explicit solution, which is stored in lookup tables. Therefore, the control allocator requires very low computational power and immediately provides an optimal solution to the torque commands issued by the flight controller. The performance of the control allocator is evaluated on a real six-rotor helicopter. Finally, based on the real experiments, the paper investigates 1) the impact of faults and 2) the influence of the reconfiguration delay on the flight stability. These experiments show that this control allocation strategy successfully increases flight performance and maintains stability of the vehicle in case of actuator(s) total failure.

***Index Terms*—Fault tolerant constrained control allocation, Parametric programming, Hexacopter, Octocopter**

I. Introduction

Today's micro aerial vehicle (MAV) systems have proven to be ready and mature enough to be used in real-world applications. These applications use MAVs as sensor carriers to easily reach spots in difficult terrain or to oversee large regions. Also, in the context of teleoperated inspection, expensive camera and stabilization systems may be integrated on the flying platform to perform inspection tasks of bridges, power lines, dam walls and other industrial facilities. [1] Flight safety is thus of primary concern. This paper focuses on failures that may occur to the propulsion groups, which constitute the highest threat to the mission. These failures can be caused by several local causes such as rotor, motor, controller or even cable failures. Therefore, it is desirable to exploit the inherent redundancy available in multirotor systems by employing fault-tolerant control algorithms.

[1]Parts of this project are supported with ECHORD TUAV funds

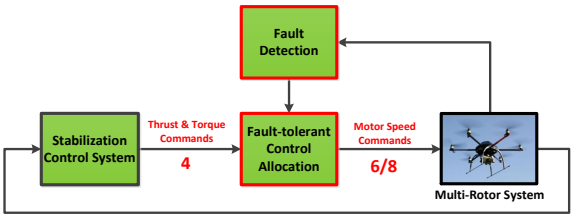


Fig. 1. Control system using fault-tolerant control allocation

In order to address actuator failures, a fault-tolerant control allocator is developed, which minimizes the flight performance degradation due to the actual fault configuration by optimally mapping the virtual controls (thrust and torques) to motor speed commands (Figure 1). The actual fault configuration is determined by a fault detection module which is not part of this work.

II. System Performance under Faults

This section describes a method to assess the static controllability of a multirotor MAV with rotors arbitrarily distributed on a plane. The method is based on the construction of the attainable control set (ACS). The ACS characterizes the capability of the MAV to produce thrust T and roll-, pitch- and yaw-torque (L, M, N) as a function of the available actuators and their physical limits. The ACS can then be used to investigate the performance degradation and to make predictions about the controllability of the system in different fault configurations.

A. Attainable Control Set

The attainable control set Φ describes a convex subspace in the thrust/torque space (TLMN) which contains all admissible

virtual control vectors $\vec{v}$. This corresponds to all thrust and torque combinations that can be produced while still satisfying the speed constraints of the motors. In this context, the ACS describes a modification of the attainable moment set (AMS) that already has been used successfully in the design of control systems for over-actuated aircrafts [1], [2], [3], [4], [5].

The space of the ACS for a system with a number of m rotors can be obtained by mapping the boundary of the m-dimensional motor constraints box Ω to in the virtual control space Φ by using the control effectiveness matrix B. In this case, the control effectiveness matrix describes the linearized relationship between the squared motor speeds $\vec{U}$ and the produced thrust/torque $\vec{v}$ as described in the next subsection. The two convex sets Φ and Ω can be defined using the following equations:

$$\Omega = \left\{ \vec{U} \in \mathbb{R}^m | \vec{U}_{\min} \leqslant \vec{U} \leqslant \vec{U}_{\max} \right\} \tag{1}$$

$$\Phi = \left\{ \vec{v} \in \mathbb{R}^n | H \cdot \vec{v} \leqslant \vec{k} \right\} \tag{2}$$

$$H \in \mathbb{R}^m,\ \vec{k} \in \mathbb{R}^{q \times n} \tag{3}$$

with $n = 4$ for a system with four control objectives as it is the case in a multirotor system with thrust and three axis torque commands. The number of constraints is q and the number of rotors is m.

The problem of finding the ACS is now to obtain the matrix H and the vector $\vec{k}$ that define the ACS as a set of inequalities. The method uses the following properties of a linear transformation: (proofs in [2])

- Convexity of a set is preserved under a linear transformation
- Points on the boundary $\partial\Phi$ of the set Φ map to unique points on the boundary $\partial\Omega$ of the set Ω
- Vectors remain parallel under linear transformation

First, the extremal vertices P_Ω of the polytope Ω are obtained

$$P_\Omega = V(\Omega) \tag{4}$$

where $V(\cdot)$ defines an operator that solves the vertex enumeration problem for a convex polytope. [6]

Second, the extremal points P_Ω are mapped to the virtual control space Φ using the control effectiveness matrix B (see the definition in II-B) to obtain a list of vertices P'_Φ

$$P'_\Phi = B \cdot P_\Omega \tag{5}$$

The points P'_Φ obtained using Eq. 5 either lie on the boundary $\partial\Phi$ or in the interior of Φ. Therefore, the list contains the extremal vertices P_Φ of the ACS Φ together with some points that are not extremal points but contained in Φ. Further, it is likely that more than one point in Ω maps to the same point in Φ as B describes as projection from a higher dimensional space onto a space of lower dimension. Because the properties of linear transformations, Φ also describes a convex polytope. A convex hull problem over the mapped vertices P'_Φ can be solved to obtain the extremal vertices P_Φ. The matrix H and the vector $\vec{k}$ that describe the polytope Φ using a set of linear inequalities, as defined in Eq. 2, can also be obtained in this way [6].

As an illustration, figure 2 shows a cut through the ACS for an hexacopter. As the ACS is a four-dimensional polytope, it has been cut at $N = 0$ (zero yaw torque) for better visualization.

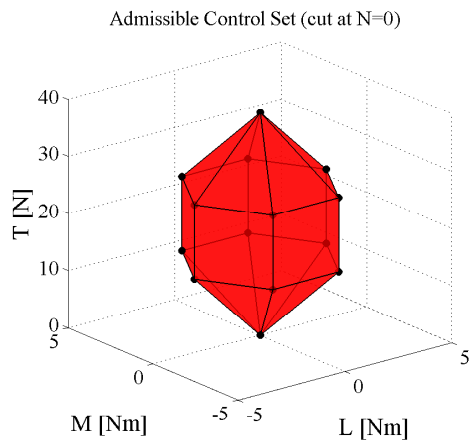


Fig. 2. ACS of an hexacopter (T: thrust, L: roll-, M: pitch-torque)

B. Control Effectiveness Matrix

The relationship between the squared rotor speed ω^2 and the thrust/torque is as follows

$$F_i = \eta_i \cdot \mu \cdot \omega_i^2 \tag{6}$$

$$T_i = -\eta_i \cdot sign(\omega_i) \cdot \kappa \cdot \omega^2 \tag{7}$$

where F_i is the thrust generated by the i-th rotor, T_i is the counter-torque of the i-th rotor and μ, κ are the aerodynamic coefficients of the rotors. The parameter η_i describes the rotor's efficiency and can be used to account for rotor wear or for a failure by setting $\eta_i = 0$.

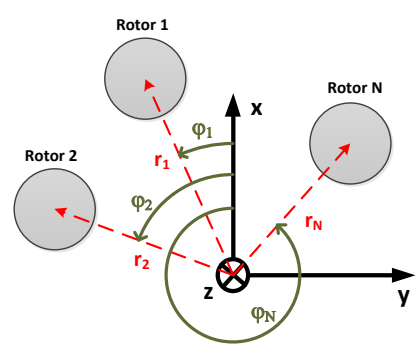


Fig. 3. Geometry definition for multirotor system

Knowing the geometry of the system (Figure 3) the control effectiveness matrix B can be obtained using the following

relationship

$$\underbrace{\begin{bmatrix} T \\ L \\ M \\ N \end{bmatrix}}_{\vec{v}} = B \cdot \underbrace{\begin{bmatrix} \omega_1^2 \\ \vdots \\ \omega_N^2 \end{bmatrix}}_{\vec{\omega}^2} \tag{8}$$

where the control effectiveness matrix B in a parametrized form is

$$B = \begin{bmatrix} t_1 & \cdots & t_N \\ l_1 & \cdots & l_N \\ m_1 & \cdots & m_N \\ n_1 & \cdots & n_N \end{bmatrix} \tag{9}$$

with the parameters defined as

$$t_i = \eta_i \cdot \mu \tag{10}$$

$$l_i = \eta_i \cdot \mu \cdot r_i \cdot \sin(\varphi_i) \tag{11}$$

$$m_i = \eta_i \cdot \mu \cdot r_i \cdot \cos(\varphi_i) \tag{12}$$

$$n_i = \eta_i \cdot \kappa \cdot d_i \tag{13}$$

and the parameters (r_i, φ_i) describe the position of the i-th rotor in the horizontal plane relative to the center of gravity in a polar coordinate system and $d_i \in \{-1, 1\}$ describes the turn direction of the i-th rotor.

C. Analysis of Static Controllability

In this section, a flight performance comparison is made in the nominal case and in different actuator-fault cases. The MAV controllability is assessed by evaluating the flight performance degradation.

The number of different fault configurations P (including fault-free case) for an helicopter with m rotors can be calculated using the following equation assuming that the order of the fault occurrence does not matter and that a minimum of four rotors have to run.

$$P = \sum_{i=0}^{m-4} \binom{m}{i} = \sum_{i=0}^{m-4} \frac{m!}{i! \cdot (m-i)!} \tag{14}$$

TABLE I
FAULT CONFIGURATIONS FOR DIFFERENT NUMBER OF ROTORS

Number of Rotors	Number of Fault Configurations
4	1
6	22
8	163
10	848

Only a few of these fault configurations are controllable and allow for a degraded thrust and torque (TLMN) control. To quantify this loss in performance, a parameter η_{LMN} is calculated, by constructing the ACS set for the faulty and the nominal cases. The two obtained four-dimensional polytopes are then cut at the nominal hovering conditions defined with $T = m_{MAV} \cdot g$ and $N = 0$ to obtain two polygons on the L-M plane. These polygons describe the set of all achievable pitch and roll torques in nominal hover flight. They have to be sufficiently large to allow a stable flight under disturbances. Figure 4 shows an example of such a cut where the two polygons are plotted above each other. The red polygon shows the ACS in the nominal case of an hexacopter, whereas the blue polygon shows the ACS in the case of one rotor has failed.

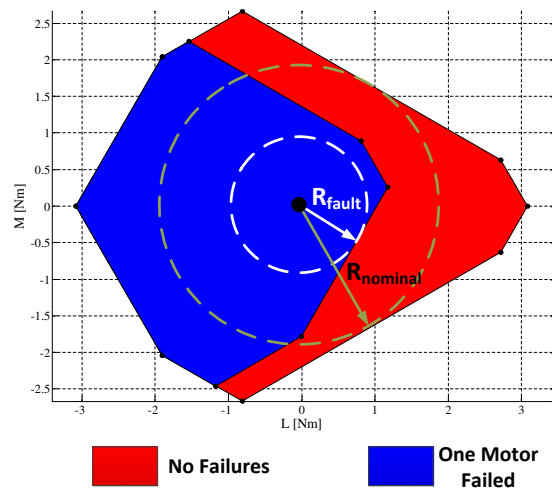


Fig. 4. Quantifying the performance degradation using cuts of the ACS

The performance degradation factor η_{LMN} can now be calculated as the ratio of the two biggest circles that fit in the polygons with the center in the origin ($L = 0$, $M = 0$).

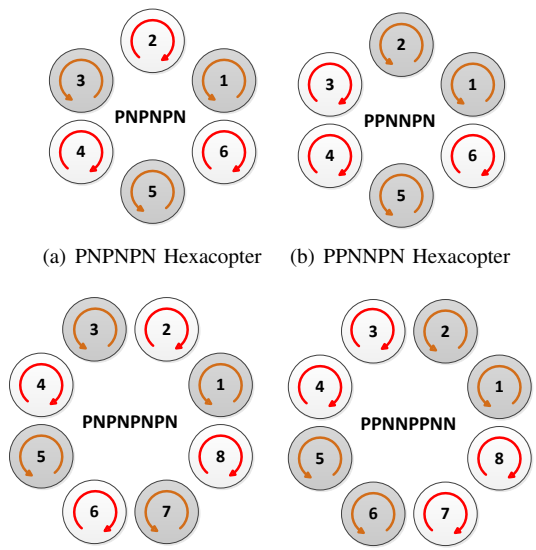
$$\eta_{LMN} = \frac{R_{fault}}{R_{nominal}} \tag{15}$$

The radius of the biggest enclosed circle expresses the minimal torque that can be produced in all directions and is an important quantity to ensure stability for arbitrary disturbances.

The controllability can be investigated by defining a minimal torque R_{min} that is needed for the MAV to operate reliably. R_{min} has to be found experimentally or can be guessed based on experience. The following condition $R_{fault} \geqslant R_{min}$ with $R_{min} > 0$ determines the controllability of a certain fault configuration. From this condition it can be seen that the vehicle is not controllable if the origin of the L-M plane is not contained in the polygon which would be equal to $R_{fault} = 0$. If the MAV under a given fault configuration is not controllable using full thrust and three torque control, it might be possible to switch to a reduced thrust, roll and pitch control without any yaw control. To assess the controllability in this case the two four-dimensional polytopes are projected onto the T-L-M hyperplane (which is orthogonal to the N coordinate) and the controllability can be assessed using the same method as described for the full TLMN control mode. In practice, however, the vehicle needs to be quite symmetrical and needs a very precise attitude estimation system to allow for a stable flight without yaw control that leads to high yaw rates. Therefore, losing the yaw control is only an option for a short period of time or as a last option in emergency situations.

D. Influence of the Number and the Arrangement of Rotors

This section compares the standard rotor arrangement of a symmetrical hexacopter and octocopter systems against a new rotor arrangement found to be best in terms of fault-tolerance. This arrangement has been found using the method described in the previous section.



(a) PNPNPN Hexacopter (b) PPNNPN Hexacopter

(c) PNPNPNPN Octocopter (d) PPNNPPNN Octocopter

Fig. 5. Traditional rotor arrangements (a,c) and arrangements for improved fault-tolerance (b,d)

Four different rotor arrangements are investigated. Figures 5(a) and 5(c) describe the most-used rotor arrangements for hexacopters and octocopters with alternating rotor-turn directions. The other two configurations, shown in Figures 5(b) and 5(d), are the arrangements that offer the best fault-tolerance capabilities among the studied multirotor systems.
For each arrangement, the values η_{LMN} and η_{LM} are calculated for all fault configurations up to a maximum of two rotors failing at the same time.
The fault-tolerance capability of different arrangements can be compared against each other by building the ratio f_{RPY} or f_{RP} that relates the number of controllable fault configurations in roll-pitch-yaw torque N_{RPY} or roll-pitch torque N_{RP} compared to the total number of possible fault configurations N_{tot}.

$$\mathrm{f}_{RPY/RP} = \frac{N_{RPY/RP}}{N_{tot}} \tag{16}$$

Table II shows these values for the four arrangements. It can be seen that the fault-tolerance capabilities of the four arrangements increase from (a) up to (d). It is interesting to note that the traditional PNPNPN hexacopter arrangement (Figure 5(a)) looses the capability of roll-pitch-yaw torque control and degrades to roll-pitch torque control as soon as only one rotor fails. However, the PPNNPN configuration (Figure 5(b)) can still maintain roll-pitch-yaw control for 33% of all fault configurations up to two random rotor failures. Moreover, 71% of these fault configurations are roll-pitch torque controllable. In the case of an octocopter using a traditional PNPNPNPN configuration, 78% of all fault configurations for two random rotor failures are still roll-pitch-yaw controllable and all fault configurations are at least roll-pitch controllable. Using the optimal PPNNPPNN arrangement the success of maintaing roll-pitch-yaw control increases to 89%.

TABLE II
FAULT-TOLERANCE COMPARISON FOR DIFFERENT MULTIROTOR ARRANGEMENTS WITH UP TO TWO ROTOR FAILURES

	Arrangement	f_{RPY}	f_{RP}	V_{ACS}
Hexacopter	(a) PNPNPN	0%	57%	145
	(b) PPNNPN	33%	71%	120
Octocopter	(c) PNPNPNPN	78%	100%	561
	(d) PPNNPPNN	89%	100%	497

Furthermore, it is relevant to note that the use of a PPNNPN instead of a PNPNPN configuration improves the fault-tolerance capabilities but also influences the geometry of the attainable control set. In the investigated cases, this led to a decrease of the ACS volume V_{ACS} which corresponds to a decreased torque-producing capability. Therefore, changing the rotor arrangement is always a trade off between fault-tolerance and performance, although the performance loss is quite small.

III. FAULT-TOLERANT CONTROL ALLOCATION

This section describes a control allocation strategy using parametric programming that is capable of exploiting the complete ACS. The overall architecture of the fault-tolerant control allocator is presented. Next in the following sections, details about the offline calculation of the explicit solution are described as well as the "constraint handler" module to ensure that the polytopic constraints on the control allocator input vectors are satisfied.

A. Architecture of the Control Allocator

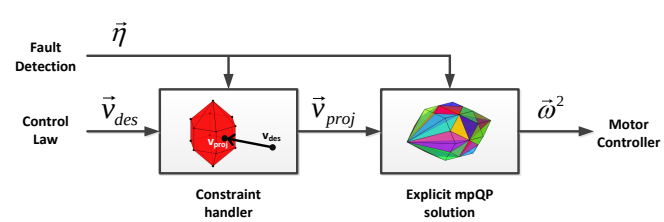


Fig. 6. Structure of the proposed control allocator

The control allocator maps the desired virtual control vector $\vec{v}_{des}$ to the motor speed commands $\vec{\omega}^2$ without violating the motor constraints. In addition, the control allocator considers the availability of the motors through the parameter vector $\vec{\eta}$. The structure of the control allocator is shown in Figure 6.
Due to the formulation of the parametric program using

hard constraints, it is necessary to ensure that the desired thrust/torque vectors $\vec{v}_{des}$ are admissible in order to correctly evaluate an explicit solution to the parametric program. The control allocator is therefore split into two subproblems to guarantee that $\vec{v}_{proj}$ always lies inside or on the surface of the attainable control set:

1) **Constraint handler:** Scales $\vec{v}_{des}$ to obtain an admissible $\vec{v}_{proj}$ which is either inside the ACS or on its surface.
2) **Explicit solution:** Uses the precomputed solution of a parametric program to solve the optimal control allocation problem. A solution only exists inside the ACS and can only be obtained with an admissible virtual control inputs $\vec{v}$.

B. Formulation as a Parametric Program

The available degree of freedom in the control allocation problem is used to minimize the sum of all rotor speeds squared $\vec{u}^T\vec{u}$ which scales with the energy consumption of the system. Hard equality constraints in the formulation ensure that the desired thrust and torque vector $\vec{v}$ always corresponds to the allocated thrust and torques. This prevents a change in direction of the commanded torques which could lead to a decrease in performance of the controlled system. The control allocation problem is formulated as a parametric program as follows

$$\begin{aligned} \vec{u}_{CA}(\vec{\theta}) &= \min_{\vec{u}} \vec{u}^T \cdot \vec{u} \\ \text{s.t. } & B \cdot \vec{u} = \vec{v} \\ & \vec{0} \leqslant \vec{u} \leqslant \text{diag}\,(\vec{\eta}) \cdot \vec{u}_{max} \end{aligned} \tag{17}$$

where $\vec{u}_{CA}(\vec{\theta})$ describes the solution to the parametric program as a function of the parameter vector $\vec{\theta}$ which is defined as

$$\vec{\theta} = \begin{bmatrix} \vec{v} & \vec{\eta} \end{bmatrix}^T \tag{18}$$

The availability of the motors is included into the parameter vector $\vec{\theta}$ using the binary subvector $\vec{\eta}$ which is defined as

$$\begin{aligned} \vec{\eta} &\in D^N \\ D &\in \{0, 1\} \end{aligned} \tag{19}$$

where the number of motors is N. In this way the motor speed constraints are parametrized to include each motor's availability.

C. Explicit Solution and Binary Search Tree

The optimal solution to the parametric program is obtained as a set of piecewise-affine (PWA) functions where each of these functions is only valid in a defined polytopic region inside the attainable control set. Therefore, it is possible to precompute all PWA functions and the corresponding region of validity. These functions can be stored in a lookup table and evaluated online to obtain the optimal solution of the control allocation problem. This is used to overcome the very limited computational resources available aboard the MAV. This way it is possible to trade computational complexity against memory which usually is available. Another advantage offered by an explicit solution is the deterministic character of the solution. This is of great importance in safety critical systems such as flying vehicles. Figure 7 shows an example of these regions in different colors for a PNPNPN hexacopter. Note that the outer hull around all these regions corresponds to the attainable control set.

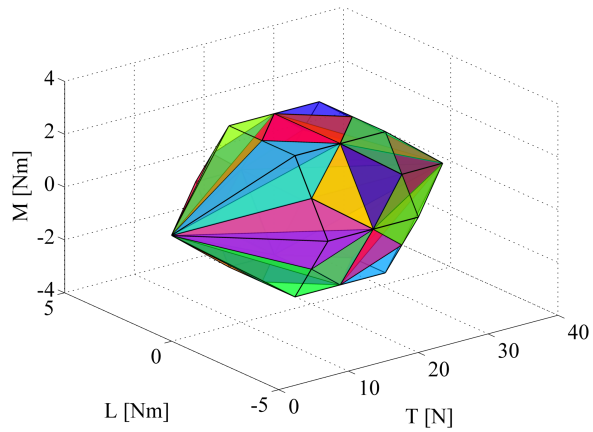


Fig. 7. Explicit solution for an PNPNPN hexacopter with 61 regions (fault-free, $N = 0$)

The control allocation problem can be solved online in two steps using this explicit solution. First, the active region has to be determined and, second, the corresponding piecewise-affine function has to be evaluated to obtain the solution. The problem of finding the active region can be solved efficiently by using a binary search tree that successively divides the attainable control set into smaller volumes using hyperplanes. This division is performed until all end nodes of the tree correspond to one region of the explicit solution. The method used to compute the search tree and the hyperplanes is then presented in [7] and [8].

The complexity of the explicit solution, however, increases drastically with the dimension of the parameter vector $\vec{\theta}$ and the number of constraints. Therefore, the complexity scales with the number of rotors. In Table III the number of regions together with the memory required to store the solution is shown for four different types of multirotor systems.

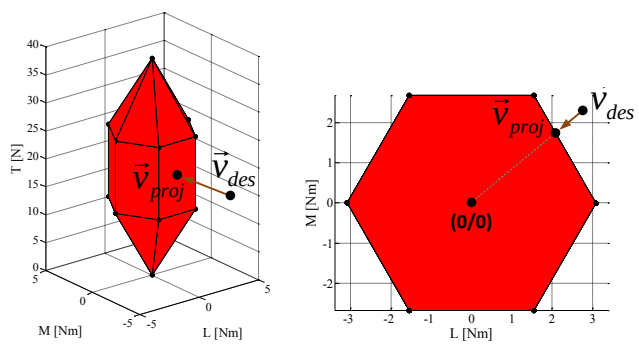
TABLE III
COMPLEXITY OF THE EXPLICIT SOLUTION

	Arrangement	Regions	Memory
Hexacopter	PNPNPN	61	40 kb
	PPNNPN	61	49 kb
Octocopter	PNPNPNPN	1049	1166 kb
	PPNNPPNN	989	1153 kb

This work uses the "Multi-Parametric Toolbox (MPT)" [10] to obtain the explicit solution of the parametric program and to calculate the binary search tree. Detailed mathematical descriptions of the calculation process can be found in [9] and [10].

D. Constraint Handler

The parametric program (Eq. 17) has no solution if the desired virtual control setpoint $\vec{v}$ lies outside the attainable control set. This is due to the use of hard constraints in the parametric program formulation. Especially in the presence of rotor failures, the virtual commands $\vec{v}$ can easily exceed the attainable control set due to the associated loss of performance. Therefore, it has to be guaranteed that such unfeasible commands are constrained before it is used as an input to the control allocator. In order to overcome this issue a "constraint handler" module is introduced between the control law and the control allocator (Figure 6). This is achieved by projecting an inadmissible $\vec{v}_{des}$ (outside the ACS) onto the surface of the ACS to obtain $\vec{v}_{proj}$ as shown in Figure 8.



(a) 3D Overview (b) Cut at $T = T_{cmd}$ and $N = N_{cmd}$

Fig. 8. Projecting of $\vec{v}_{des}$ onto the surface of the attainable control set Φ

Such a projection preserves the direction of the resulting torques $\vec{v}_{proj}$. The desired control command is thus only scaled in magnitude. In order to be able to project an inadmissible $\vec{v}_{des}$ onto the surface for all possible fault configurations, the attainable control set has to be precomputed offline and stored in lookup tables for all cases. The attainable control sets Φ_i are stored in its half-space representation as a set of inequalities for each fault configuration

$$\Phi_i = \left\{ \vec{v} \in \mathbb{R}^n | H_i \cdot \vec{v} \leqslant \vec{k}_i \right\} \tag{20}$$

$$H_i = \begin{bmatrix} \vec{h}_1^T \\ \vdots \\ \vec{h}_M^T \end{bmatrix}, \ \vec{k}_i = \begin{bmatrix} k_1 \\ \vdots \\ k_M \end{bmatrix} \tag{21}$$

where H_i and $\vec{k}_i$ define the ACS for the i-th fault configuration, M is the number of half-spaces that define the ACS, $\vec{h}_j$ the normal vector corresponding to the j-th half-space and k_j the position of the half-space defined as

$$\vec{h}_i \cdot \vec{v} \leqslant k_i \tag{22}$$

The projection $\vec{v}_{proj}$ can then be calculated using the following steps where the following point C is used as the projection center. This leaves the thrust unchanged and scales only the torque subvector in length with T_{des} being the commanded thrust.

$$C = \begin{bmatrix} T_{des} & 0 & 0 & 0 \end{bmatrix}^T \tag{23}$$

First, the intersection of the half-line starting from C to $\vec{v}_{des}$ is calculated for all hyperplanes that enclose the ACS. For the j-th hyperplane this can be achieved using the following two equations:

$$t_j = \frac{k_j - h_j^T \cdot C}{\vec{h}_j^T \cdot (\vec{v}_{des} - C)} \tag{24}$$

$$I_j = C + (\vec{v}_{des} - C) \cdot t_j \tag{25}$$

where I_j is the intersection point with the j-th hyperplane. Two conditions are used to find $\vec{v}_{proj}$ among all the intersection points. First, t_j has to be positive as we are looking for an intersection of a hyperplane and a half-line with an associated direction. Second, the distance between C and $\vec{v}_{proj}$ has to be minimal which is equal to the minimal value of t_j, $\forall j \in [1..M]$. This can be written as

$$t_{proj} = \min \left(\begin{bmatrix} t_1 & \cdots & t_M \end{bmatrix} \right), \ \mathrm{t} \geqslant 0 \tag{26}$$

the projection of $\vec{v}_{des}$ is then obtained with

$$\vec{v}_{proj} = C + (\vec{v}_{des} - C) \cdot t_{proj} \tag{27}$$

E. Properties of the Control Allocator

The widely used pseudo-inverse (PSI) allocator is not able to allocate controls for the entire ACS. A major advantage of the proposed control allocator is the capability of using the entire ACS, which leads to an increase in control performance, since higher torques can be achieved with the same physical system. This is possible because the ACS is divided into regions for each of which a specific relationship between $\vec{v}$ and $\vec{\omega}^2$ exists. As a measure of performance for the control allocators the ratio r between the volume of the ACS and the volume of the allocatable subspace is defined as

$$r = \frac{V_{PSI}}{V_{ACS}} \tag{28}$$

Table IV lists this ratio for four types of multirotor systems. Figure 9 shows the allocatable subspaces for the PSI as well as the mpQP allocator for a hexacopter with a PNPNPN configuration.

TABLE IV
r VALUES FOR THE PSI AND MPQP ALLOCATORS

	Arrangement	r_{PSI}	r_{mpQP}
Hexacopter	PNPNPN	50 %	100 %
	PPNNPN	52 %	
Octocopter	PNPNPNPN	34 %	
	PPNNPPNN	43 %	

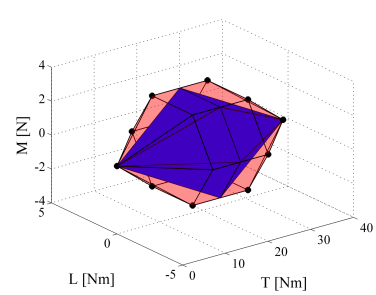
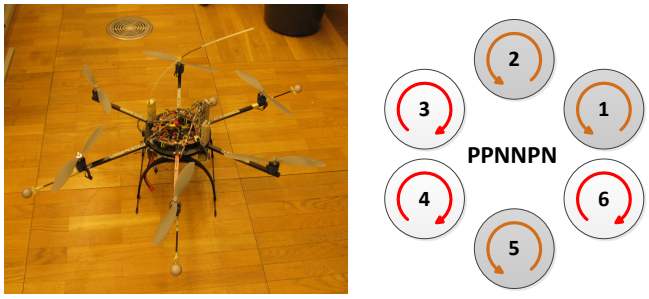


Fig. 9. Allocatable space comparison (red:mpQP, blue: PSI)

IV. Experiments

An indoor test environment has been created to test and compare the behavior and performance of the designed control allocator on a real hexacopter.

A. Setup and Environment



(a) Hexacopter system (b) Rotor arrangement

Fig. 10. Hexacopter used for the experiments provided by Skybotix AG

The used MAV is an hexacopter using a PPNNPN rotor arrangement as shown in Figure 10(b). A Vicon visual motion-tracking system is used to obtain very precise position and angle information at a rate of 200 Hz. This information is used to evaluate the performance of the control allocator as well as to drive a position controller to keep the MAV at a defined location during the injection of rotor failures. A video showing the fault recovery process using the proposed control allocator is available on the internet. [2].

B. Comparison mpQP / Pseudo-Inverse Control Allocator

The performance of the traditional pseudo-inverse allocator is compared against the new parametric programming control allocator. The hexacopter is brought to a stable hovering position using the position and height controller. At this position, motor 3 is turned off and the control allocator is reconfigured after a delay of $500\ ms$. As the position controller commands non-zero setpoints to the attitude controller, the Euler angle tracking errors are used to evaluate the experiments.

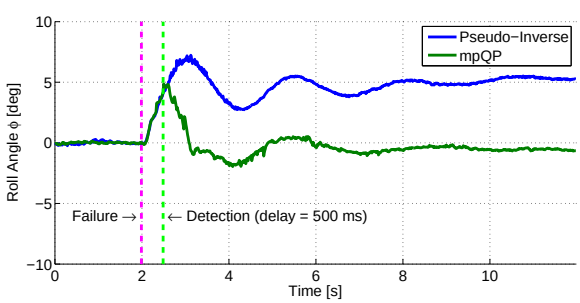


Fig. 11. Tracking error of roll angle ϕ for the two control allocators

Looking at the roll tracking error in Figure 11, it can be seen that the error quickly grows after the motor has failed at $t = 2\ s$. After the reconfiguration of the control allocator at $t = 500\ ms$, it can be seen that the error quickly decreases back to nominal values for the parametric programming allocator. The pseudo-inverse allocator, however, leads to a steady-state error of around $5\ deg$ in the roll angle and is therefore not able to recover from the fault completely. Although the tracking errors remain bounded, the introduced offsets lead to high position errors very quickly, especially in situations with poor position information where such offsets can only be estimated poorly.

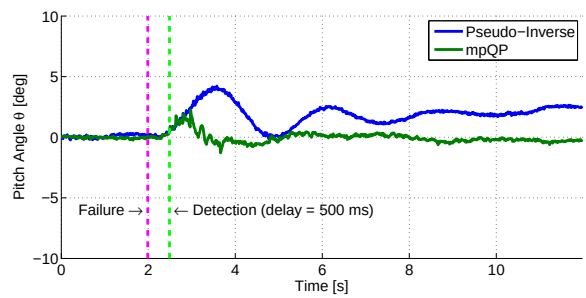


Fig. 12. Tracking error of pitch angle θ for the two control allocators

C. Influence of the Reconfiguration Delay

The effect of the reconfiguration delay on the angle tracking error is investigated in this experiment using the proposed parametric programming control allocator. The reconfiguration delay in this context describes the time between the actual motor failure and the time at which the control allocator is reconfigured. Again, the hexacopter is brought to stable hover flight at a defined position in the room using the horizontal and height controllers. In stable hovering, motor 3 is turned off and after the reconfiguration delay has elapsed, the control allocator is reconfigured to account for the failure. The roll tracking error is shown in Figure 13 for three different reconfiguration delays.

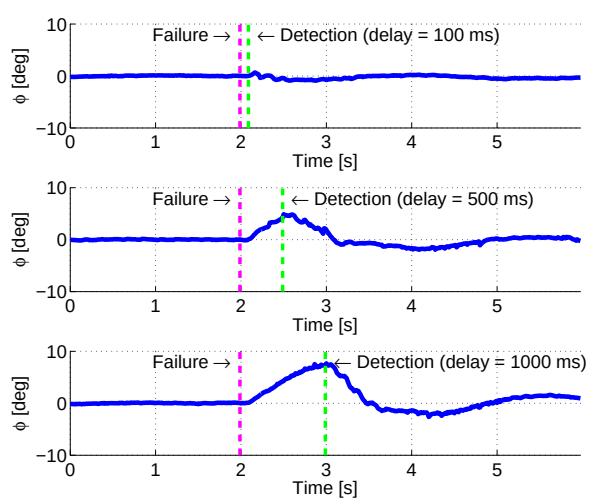


Fig. 13. Roll tracking error with different reconfiguration delays

[2]http://www.youtube.com/watch?v=CIuEv-SuBnQ

It can be seen in Figure 13 that the smaller the reconfiguration delay, the smaller the impact of the motor fault on the vehicle's attitude. Thanks to the new proposed control allocator, the attitude of the MAV is stabilized quickly and accurately after a fault occurrence despite a long detection delay. This leaves a significant amount of time for a fault detection and isolation (FDI) system to diagnose accurately the origin of the fault, thus limiting the risk of false failure detection and control allocation reconfiguration.

V. Conclusion

A method has been proposed to assess the static controllability of multirotor systems under propulsion group failures based on the construction of the attainable control set. Using this method it has been shown that the rotor arrangement has a big influence on the fault-tolerance capabilities of the system. A fault-tolerant control allocator based on the theory of parametric programming has been developed and an explicit solution together with a binary search tree is obtained using the "Multi-Parametric Toolbox" [10]. In addition, a constraint handler module is introduced to scale inadmissible torques commanded by the control law by projecting the control vector onto the surface of the attainable control set to preserve the direction of the commanded torques. The volume of admissible virtual controls has been increased by around 50% to 66% for an hexacopter using the new control allocator. This corresponds to the ability of producing higher torques using the same phsysical system.

The new allocator was implemented on a real hexacopter and the performance was evaluated. These tests showed a good behavior in fault recovery. Additionally, it has been shown that the reconfiguration delay can easily be as high as $1\ s$ without negative effects on the platform's stability. This is particularly useful for a reliable fault diagnosis.

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